

Analysis First-Year Exam, May 2020, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $a < b$ and let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable. Prove that if $\int_a^b f = 0$, then for each positive integer n there exist $c < d$ in $[a, b]$ such that $\sup\{f(t) : c \leq t \leq d\} < 1/n$. Hence, or otherwise, prove that if $f > 0$ throughout $[a, b]$, then $\int_a^b f > 0$.
2. Let $(x_n : n \geq 0)$ converge in the metric space X and let $(f_n : n \geq 0)$ be a sequence of continuous real-valued functions on X .
 - (a) Prove that if the sequence of functions is uniformly convergent, then $(f_n(x_n) : n \geq 0)$ is convergent.
 - (b) Show by example that if uniformly is replaced by pointwise, then $(f_n(x_n) : n \geq 0)$ may not converge.
3. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and assume that

$$\int_0^1 f(t)t^n dt = 1/(n+1)$$

for each integer $n > 1$. Deduce as much as is possible about f .

4. Let $(f_n : n \geq 0)$ be a sequence of measurable real-valued functions on some measurable space. Prove that each of the following sets is measurable:

$$A = \{\omega : f_n(\omega) \in [0, 1] \text{ for finitely many } n\}$$

$$B = \{\omega : f_n(\omega) \in [0, 1] \text{ for infinitely many } n\}$$

$$C = \left\{ \omega : \text{the series } \sum_{n=0}^{\infty} f_n(\omega) \text{ is absolutely convergent} \right\}.$$

5. For each positive integer n , let $f_n : [0, 1] \rightarrow [0, 1]$ be continuous; assume that $f_n \rightarrow 0$ pointwise as $n \rightarrow \infty$. Does it follow that

$$\int_0^1 f_n(t) dt \rightarrow 0 \quad \text{as } n \rightarrow \infty?$$

Does your answer change if continuous is replaced by Riemann integrable?